Convergent Evidence for 5×5 Polybius Structure in Kryptos K4

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Abstract

Two independent observations about Kryptos K4 converge on a single structure: the Kryptos Alphabet arranged as a 5-wide Polybius grid. (1) The 17 consensus null positions contain only letters from columns 0 and 3 of this grid ($p \approx 6 \times 10^{-5}$, Monte Carlo). (2) The Beaufort keystream at all 24 crib positions is enriched for the same column-$\{0, 3\}$ letters ($p = 4.3 \times 10^{-3}$, cross-validated on disjoint position sets). Direct simulation of both observations jointly yields $p = 1.4 \times 10^{-7}$ ($N = 50{,}000{,}000$ letter permutations). A third observation—Bean (2021) found that 13/24 keystream values are multiples of 5 in the reversed KA indexing ($p \approx 9.5 \times 10^{-4}$)—shares a mod-5 connection with the grid but does not cleanly reduce to a column selection (Section 4). All three observations are model-independent: they depend only on the carved ciphertext and the known cribs, not on any cipher model, keyword, or null mask hypothesis.

1 Setup

Let $\mathcal{A} = \{A, B, \ldots, Z\}$ with standard indexing A=0, …, Z=25. All arithmetic modulo 26.

Definition 1 (Kryptos Alphabet)**.** KA = *KRYPTOSABCDEFGHIJLMNQUVWXZ, with* KA[0]=K, KA[1]=R, …, KA[25]=
All 26 letters appear exactly once.

Definition 2 (K4 Ciphertext and Cribs)**.**

$$C = \texttt{OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJQSSEKZZWAT}$$
$$\texttt{JKLUDIAWINFBNYPVTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR}$$

with $|C| = 97$ *and 0-indexed positions. Known cribs (24 characters total):*

$$P[21..33] = \texttt{EASTNORTHEAST} \quad (13\ chars),$$
$$P[63..73] = \texttt{BERLINCLOCK} \quad (11\ chars).$$

Definition 3 (5-Wide KA Polybius Grid)**.** *Arrange KA into a grid with 5 columns:*

	c_0	c_1	c_2	c_3	c_4
r_0	**K**	*R*	*Y*	*P*	*T*
r_1	**O**	*S*	*A*	**B**	*C*
r_2	*D*	*E*	*F*	**G**	*H*
r_3	**I**	*J*	*L*	*M*	*N*
r_4	*Q*	*U*	*V*	**W**	*X*
r_5	**Z**				

where bold letters occupy columns 0 and 3 exclusively. For letter ℓ, define $\mathrm{col}(\ell) = \text{KA-idx}(\ell) \bmod 5$.

Definition 4 (Palette)**.** $\Pi = \{\ell \in \mathcal{A} : \mathrm{col}(\ell) \in \{0, 3\}\} = \{B, G, I, K, O, W, Z\}$.

Definition 5 (Consensus Null Positions)**.** *The 17 positions identified with 100% agreement across 6 independently optimized null masks (each maximizing crib consistency via simulated annealing):*

$$\mathcal{N}_{17} = \{0, 1, 2, 5, 8, 12, 14, 20, 36, 52, 58, 59, 74, 75, 78, 84, 85\}.$$

2 Proposition 1: Null Palette Restriction

Proposition 1 (Null Palette)**.** *The characters at the 17 consensus null positions use only letters from* Π*:*

$$\{C[i] : i \in \mathcal{N}_{17}\} = \{B, G, I, K, O, W, Z\} = \Pi.$$

Proof. Direct enumeration. The 17 characters are $O, B, K, O, G, B, O, W, W, K, W, I, W, G, Z, I, G$, with multiplicities $B^2 G^3 I^2 K^2 O^3 W^4 Z^1$. All 7 letters belong to Π, and conversely every element of Π appears at least once. □

Statistical test. Under H_0, the 17 null positions are a random subset of the 73 non-crib positions in C. We estimate $P(\text{distinct letters} \leq 7)$ by Monte Carlo ($N = 2{,}000{,}000$ trials, sampling 17 from 73 without replacement):

$$p = 6.3 \times 10^{-5}.$$

Expected distinct count: 12.41 ± 1.32; observed: 7; $z = -4.09$.

Post-hoc status. The palette Π was identified from the data. The Monte Carlo p-value tests the well-defined statistic "number of distinct letters in 17 randomly chosen non-crib positions," which is the appropriate post-hoc test. The specific identity of the 7 letters is a separate, unquantified observation.

Circularity defense. The 17 positions were identified by SA optimization of *positional alignment* (which positions to remove so that cribs land correctly). The palette is a property of which *letters* sit at those positions—an unoptimized quantity. Position selection depends on alignment geometry, not on letter identity.

3 Proposition 2: Beaufort Keystream Enrichment

Definition 6 (Beaufort Keystream at Crib Positions)**.** *For each known crib position i:*

$$k[i] = (C[i] + P[i]) \bmod 26.$$

This value is ***model-independent****: it depends only on the carved text and the known plaintext, not on any cipher model, keyword, or null mask.*

Proposition 2 (Cross-Validated Palette Enrichment)**.** *Of the 24 Beaufort keystream values at crib positions, 13 are members of* Π*:*

$$|\{i \in \mathcal{C} : k[i] \in \Pi\}| = 13.$$

Proof. Direct computation at all 24 crib positions.

EASTNORTHEAST (positions 21–33):

Pos	$C[i]$	$P[i]$	$k[i]$	$\in \Pi$?
21	F	E	J (9)	No
22	L	A	L (11)	No
23	R	S	J (9)	No
24	V	T	O (14)	Yes
25	Q	N	D (3)	No
26	Q	O	E (4)	No
27	P	R	G (6)	Yes
28	R	T	K (10)	Yes
29	N	H	U (20)	No
30	G	E	K (10)	Yes
31	K	A	K (10)	Yes
32	S	S	K (10)	Yes
33	S	T	L (11)	No

ENE palette hits: 6/13.

BERLINCLOCK (positions 63–73):

Pos	$C[i]$	$P[i]$	$k[i]$	$\in \Pi$?
63	N	B	O (14)	Yes
64	Y	E	C (2)	No
65	P	R	G (6)	Yes
66	V	L	G (6)	Yes
67	T	I	B (1)	Yes
68	T	N	G (6)	Yes
69	M	C	O (14)	Yes
70	Z	L	K (10)	Yes
71	F	O	T (19)	No
72	P	C	R (17)	No
73	K	K	U (20)	No

BCL palette hits: 7/11. Total: 13/24. □

Statistical test. Under H_0 (keystream uniform over $\mathcal{A}$), $P(\text{palette letter}) = |\Pi|/26 = 7/26 \approx 0.269$.

$$P(X \geq 13 \mid n{=}24,\ p{=}7/26) = 4.3 \times 10^{-3}.$$

Cross-validation. The palette Π was derived from positions $\mathcal{N}_{17} = \{0, 1, 2, 5, 8, \ldots, 85\}$. The enrichment is measured at crib positions $\mathcal{C} = \{21, \ldots, 33, 63, \ldots, 73\}$. These sets are **strictly disjoint**: $\mathcal{N}_{17} \cap \mathcal{C} = \emptyset$. This constitutes genuine out-of-sample validation: the palette definition and the enrichment test share zero data points.

Variant specificity. Only Beaufort ($A{=}0$) achieves 13/24. Vigenère gives 9/24; Variant Beaufort gives 5/24.

4 Bean's Mod-5 Observation (2021)

Proposition 3 (Bean's Mod-5 Observation)**.** *Define the reversed Kryptos Alphabet* rKA = `ZXWVUQNMLJIHGFEDCBAS` *and compute:*

$$\beta[i] = (\text{rKA-idx}(C[i]) - \text{idx}(P[i])) \bmod 26$$

at each crib position i*. Then* 13 *of* 24 *values satisfy* $\beta[i] \equiv 0 \pmod 5$*.*

Proof. Bean (2021, footnote 3). We independently verify: $P(X \geq 13 \mid n{=}24,\ p{=}6/26) = 9.5 \times 10^{-4}$. □

Apparent structural connection (incorrect). An earlier draft of this paper claimed that $\beta[i] \equiv 0 \pmod 5$ is equivalent to $\text{col}(C[i]) \equiv -\text{idx}(P[i]) \pmod 5$, which would directly select KA Polybius columns. This equivalence is **wrong**: it requires $(\text{KA-idx}(C[i]) + \text{idx}(P[i])) \bmod 26 \equiv (\text{KA-idx}(C[i]) + \text{idx}(P[i])) \bmod 5$, which holds only when the intermediate sum is less than 26. Since $26 \equiv 1 \pmod 5$, reducing mod 26 before mod 5 introduces a ± 1 shift for values ≥ 26. Empirically, only 9 of the 13 Bean mod-5 hits satisfy the column congruence.

Relationship to the Polybius grid. Bean's mod-5 and the palette enrichment share a surface connection: both involve divisibility by 5, and the KA Polybius grid has 5 columns. However, the algebraic reduction from Bean's statistic to column selection does not hold exactly. Bean's observation is a genuine, independently significant anomaly ($p = 9.5 \times 10^{-4}$) that is *related to* but not *equivalent to* the Polybius column structure.

Independence. Bean's observation was published in 2021, before the palette Π was identified. The two findings were made by different researchers, from different starting points (reversed-KA keystream arithmetic vs. null-character identity). Whether the shared mod-5 structure reflects a common mechanism or a coincidence of the arithmetic remains an open question.

5 K-Sum Bias

Proposition 4 (Keystream K-Sum Deficit)**.** *The Beaufort keystream at the 24 crib positions is* `JLJODEGKUKKKLOCGGBGOKTRU`*. Mapping each letter to its KA index and summing:*

$$\Sigma = \sum_{i \in \mathcal{C}} \text{KA-idx}(k[i]) = 218.$$

Under a uniform keystream, $E[\Sigma] = 24 \cdot 25/2 = 300$, $\text{SD}(\Sigma) = \sqrt{24 \cdot (26^2 - 1)/12} \approx 36.7$*. The deficit is* 82*, giving* $z = -2.23$ *(*$p \approx 0.013$*, one-tailed).*

Interpretation. A low KA-sum means the keystream is biased toward *early* KA letters—i.e., the keyword `KRYPTOS` and its immediate successors. Indeed, 10/24 keystream letters (42%) are in $\{K, R, Y, P, T, O, S\}$ versus 27% expected (7/26). This is a fourth signal pointing to KA structure in the keystream.

Conditional status. The K-sum deficit is largely a consequence of the palette enrichment (Proposition 2): palette letters Π occupy early KA positions (mean KA-index 7.7 vs. 12.5 overall), so any keystream enriched for Π will mechanically produce a low KA-sum. The K-sum is therefore a *conditional redescription* of the palette signal, not an independent fourth observation.

Note. This observation was first noted by Bean in private correspondence (March 2026), who reported the K-sum as "the only interesting thing about that key."

6 Combined Evidence

Theorem 1 (Cross-Validated Polybius Structure)**.** *Propositions 1 and 2 are measured on strictly disjoint position sets ($\mathcal{N}_{17}$ vs. $\mathcal{C}$). However, both statistics are deterministic functions of the same 97-character ciphertext, so Fisher's method (which assumes independence) is not formally valid.*

We instead compute the joint significance by direct simulation. In $N = 50{,}000{,}000$ random permutations of the 97 CT letters (seed 20260320), we re-derive the palette from the null positions of each shuffled text and test keystream enrichment for that derived palette:

$$P\big(\mathit{distinct} \le 7 \ \wedge\ \mathit{ks\_palette} \ge 13 \mid \mathit{shuffled\ CT}\big) = \frac{7}{50{,}000{,}000} = 1.4 \times 10^{-7},$$

with 95% CI $[6.8 \times 10^{-8},\ 2.9 \times 10^{-7}]$.

Among the 1,405 shuffles that produced ≤ 7 distinct null letters, the mean keystream enrichment for the derived palette was 6.64/24 (random expectation: 6.46). Only $7/1{,}405 = 0.50\%$ also achieved $\ge 13/24$ keystream enrichment. This confirms that palette restriction does not automatically produce keystream enrichment—the cross-validation is genuine.

Interpretation. Two properties of the K4 ciphertext—the letter identity at null positions and the Beaufort keystream at crib positions—both select for the same 7-letter subset defined by columns 0 and 3 of the 5-wide KA Polybius grid, with joint significance $p \approx 1.4 \times 10^{-7}$ under letter-permutation null.

Note on Fisher's method. The Fisher combined $p = 4.2 \times 10^{-6}$ reported in an earlier draft of this paper is not valid because the component statistics share the same ciphertext realization. The direct simulation above yields a *stronger* result (1.4×10^{-7} vs. 4.2×10^{-6}) because the two statistics are negatively correlated under the null: shuffled texts that produce restricted palettes tend *not* to enrich keystreams for those palettes.

Relationship to Bean's mod-5. Bean's mod-5 observation (Proposition 3, $p = 9.5 \times 10^{-4}$) was made independently in 2021 by a different researcher using a different computation. As discussed in Section 4, Bean's statistic shares a mod-5 connection with the Polybius grid but does not reduce cleanly to a column selection. The primary convergence is *two-way*: null palette restriction and keystream enrichment both select the same 7-letter subset defined by the 5-wide KA grid. Bean's observation is a related but structurally distinct signal.

7 Pre-Registered vs. Post-Hoc Status

Observation	Status	Note
Palette $\lvert\Pi\rvert \le 7$	Post-hoc	MC p valid for ≤ 7 distinct
Palette $= \{B, G, I, K, O, W, Z\}$	Post-hoc	Specific letters from data
Keystream 13/24 palette	**Cross-validated**	Disjoint position sets
Beaufort $>$ other variants	Bonferroni-corrected	3 variants tested
Bean mod-5 (2021)	Independent	Related but not equivalent to grid
K-sum = 218	*Conditional*	Driven by palette enrichment

8 Open Question

Bean (2021) reported two further observations on the raw 97-character ciphertext: (a) width-21 vertical bigram repeats (11/76, $p \approx 1/6{,}750$) and (b) same-plaintext-letter ciphertext-distance clustering ($p \approx 1/240$). Both depend on non-crib characters.

Question 1. Do these signals survive on the 73-character text after removing the 17 consensus null positions? The mod-5 signal is invariant (it depends only on crib positions, which are never nulls). But width-21 and proximity observations use non-crib characters whose positions shift after null extraction. If they weaken, the stego layer was generating them. If they persist, they constrain the cipher layer.

Question 2. The palette enrichment (Proposition 2) is measured on the raw 97-character carved text. Under the best-known cipher model (null mask + columnar transposition + Beaufort), the keystream operates in a transposed intermediate space where palette enrichment drops to 5/24—*below* the random expectation of 6.46. This does not invalidate the raw-CT97 observation (which is model-independent), but raises the question: does the Polybius signal exist in the carved text as an artifact of the stego layer, or does it survive into the cipher layer under the correct (still unknown) decryption model?

Reproducibility

All computations use the `kryptos` kernel (`src/kryptos/kernel/constants.py`, which self-verifies CT, cribs, and Bean constraints at import time). Palette Monte Carlo: seed `20260320`, $N = 50{,}000{,}000$ trials. Joint null simulation: `scripts/statistical/e_joint_null_simulation.py`, seed `20260320`, $N = 50{,}000{,}000$ trials per model. Source code: `github.com/jcolinpatrick/kryptos`.

References

[1] R. Bean, "Cryptodiagnosis of Kryptos K4," *Proceedings of the 4th International Conference on Historical Cryptology (HistoCrypt 2021)*, pp. 13–21.

[2] J. Sanborn, Dedication speech, November 3, 1990. RR Auction lot documentation.